

Multi-Granularity Multi-Modal Knowledge Graph Representation Learning via Subgraph-Aware Adaptive Fusion and Hierarchical Relation Modeling

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Abstract

Multi-modal knowledge graphs (MMKGs) enrich traditional knowledge graphs by incorporating heterogeneous modalities such as textual descriptions and visual content, offering complementary semantic cues for knowledge reasoning. However, existing approaches often overlook the structural dependencies within each modality, apply static or coarse-grained fusion strategies, and insufficiently model relational semantics. We propose a Multi-Granularity Multi-Modal Knowledge Graph Representation Learning Method via Subgraph-aware Addaptive Fusion and Hierarchical Relation Modeling (SAFER), which implement multi-modal knowledge representation through adaptive fusion of multi-granularity information such as multi-modal semantics, knowledge structures and relations. SAFER explicitly constructs modality-specific subgraphs and employs structure-aware graph attention networks to effectively capture intra-modal structural dependencies. We propose an adaptive multi-modal fusion mechanism, which aggregates modality-specific embeddings at the semantic level by dynamically assigning entity-specific modality weights. We design a two-stage multi-granularity knowledge relation modeling strategy, which utilizes a structure-aware multi-modal adaptive pre-fusion to preserve

topological information and a relation-aware graph attention network (RGAT) post-fusion to encode relational semantics. Extensive experiments on several benchmark datasets demonstrate that the proposed SAFER significantly outperforms competitive baselines on link prediction and relation reasoning tasks.

CCS Concepts

• Information systems → Multimedia information systems.

Keywords

Multi-Modal Knowledge Graph, Representation Learning, Modality-Aware GNN, Relation-Aware Graph Attention, Knowledge Completion, Multi-Grained Reasoning

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1 Introduction

Knowledge Graphs (KGs) have emerged as a fundamental framework to represent structured knowledge in various artificial intelligence applications, including logic reasoning [40, 42], recommendation systems [33], and question answering [7]. Typically, KGs store knowledge as relational triples (*head, relation, tail*), encoding entities and their semantic interactions explicitly. To enable effective reasoning and inference over these symbolic structures,

Knowledge Graph Embedding (KGE) techniques have been extensively studied to map entities and relations into continuous vector spaces, effectively capturing latent semantic structures. Early KGE methods, such as TransE [2], DistMult [31], and RotatE [22], have demonstrated promising performance by learning embeddings solely based on graph structural information. However, due to their dependence on structural connectivity alone, these approaches face inherent limitations in capturing richer semantic context, especially in scenarios characterized by data sparsity or incomplete knowledge.

To address these shortcomings, recent advancements have increasingly focused on integrating multi-modal information into KGs, leading to the emergence of Multimodal Knowledge Graphs (MMKGs) [17, 20, 29]. Unlike conventional symbolic KGs, MMKGs enrich entity representations with diverse external modalities such as textual descriptions, images, videos, and even audio, providing complementary semantic cues that significantly enhance the quality and comprehensiveness of knowledge representation. A variety of multi-modal embedding methods, including IKRL [29], TBKGC [18], and MoSE [39], have been proposed to exploit such multi-modal signals, improving knowledge graph representation and facilitating more accurate downstream reasoning. These approaches typically perform modality fusion at either the structural or semantic level, attempting to unify heterogeneous information from various sources into coherent entity embeddings.

Despite recent advances, existing multi-modal knowledge representation methods still face several limitations. (1) Challenge 1: Most methods do not explicitly construct modality-specific subgraphs, directly merging features or triples without adequately modeling intra-modal structures. This neglects inherent structural patterns within each modality and may cause cross-modal interference. (2) Challenge 2: Many existing multi-modal fusion strategies adopt an early fusion strategy, aggregating modalities prematurely without sufficiently capturing inter-modal correlations, leading to semantic conflicts and information loss. (3) Challenge 3: Existing methods inadequately distinguish fine-grained relational semantics, treating relationships uniformly and thus limiting the discriminative power in representing nuanced interactions.

To address these challenges, we propose a novel Multi-Granularity Multi-Modal Knowledge Graph Representation Learning Method via Subgraph-aware Addaptive Fusion and Hierarchical Relation Modeling (**SAFER**). (1) To preserve intra-modal structure and avoid modality interference (Challenge 1), SAFER constructs modality-specific subgraphs and applies Graph Attention Networks (GAT) to independently encode structural patterns within each modality. (2) To prevent premature integration and better capture cross-modal dependencies (Challenge 2), a structure-aware multi-modal self-adaptive semantic-level fusion mechanism is proposed, which adaptively assigns entity-specific modality weights to dynamically aggregate multi-modal semantic features. (3) To enhance relational reasoning (Challenge 3), we design a two-stage multi-granularity relation modeling strategy, a structure-aware encoder encodes proximity before fusion, followed by a relation-aware RGAT that incorporates explicit relation types after fusion.

In summary, the main contributions of our work are as follows:

- we propose a novel Multi-Granularity Multi-Modal Knowledge Graph Representation Learning Method via Subgraph-aware Addaptive Fusion and Hierarchical Relation Modeling (**SAFER**), which is a unified multi-modal knowledge representation framework that jointly models modality-specific structure, multi-modal adaptive semantic fusion, and multi-granularity relation reasoning. By decoupling multi-modal entity encoding, fusion, and relational modeling, SAFER effectively addresses structural inconsistency, modality imbalance, and semantic under-specification.
- We propose a structure-aware multi-modal adaptive fusion mechanism at the semantic level, which constructs modality-specific subgraphs to model intra-modal structural dependencies based on graph attention networks, and dynamically aggregates embeddings across modalities by learning entity-specific modality contribution weights, enabling unified semantic representation.
- We design a two-stage multi-granularity relation modeling strategy that separately encodes structural proximity and explicit relational semantics, enhancing the expressiveness and discriminative power of the learned unified multi-modal knowledge representations.
- Extensive experiments on several benchmark datasets show that the proposed SAFER consistently outperforms state-of-the-art baselines, demonstrating robustness under modality sparsity and the hierarchical complexity of graph relations.

2 Related Work

2.1 Knowledge Graph Representation Learning

Knowledge graph representation learning aims to embed entities and relations into low-dimensional continuous vector spaces while preserving the structural and semantic information inherent in the graph.

Structural Embedding Methods. Early approaches focus on modeling symbolic triples in knowledge graphs. TransE [2] represents relations as vector translations between entities. DistMult [31] employs a bilinear scoring function with a diagonal relation matrix, while ComplEx [24] extends it to complex space to handle asymmetric relations. RotatE [22] interprets relations as rotations in the complex plane, and PairRE [5] introduces dual transformation vectors for head and tail entities. Tucker [1] adopts Tucker decomposition to model high-order interactions among entities and relations. These models are effective in learning structural patterns but are limited by their reliance solely on symbolic information.

Multimodal Embedding Methods. To enrich entity representations, recent works incorporate auxiliary modalities such as text and images. IKRL [29] integrates visual features via attention to complement structural embeddings. TBKGC [18] and TransAE [28] extend translation-based methods with multimodal fusion mechanisms. MMKRL [17] proposes alignment constraints and adversarial training to robustly integrate multiple modalities. LAFA [20] introduces link-aware weighting to filter irrelevant visual content, while RSME [26] designs relational-aware semantic filtering to align structure and semantics. MoSE [39] decouples modality-specific learning and dynamically combines predictions through ensemble inference.

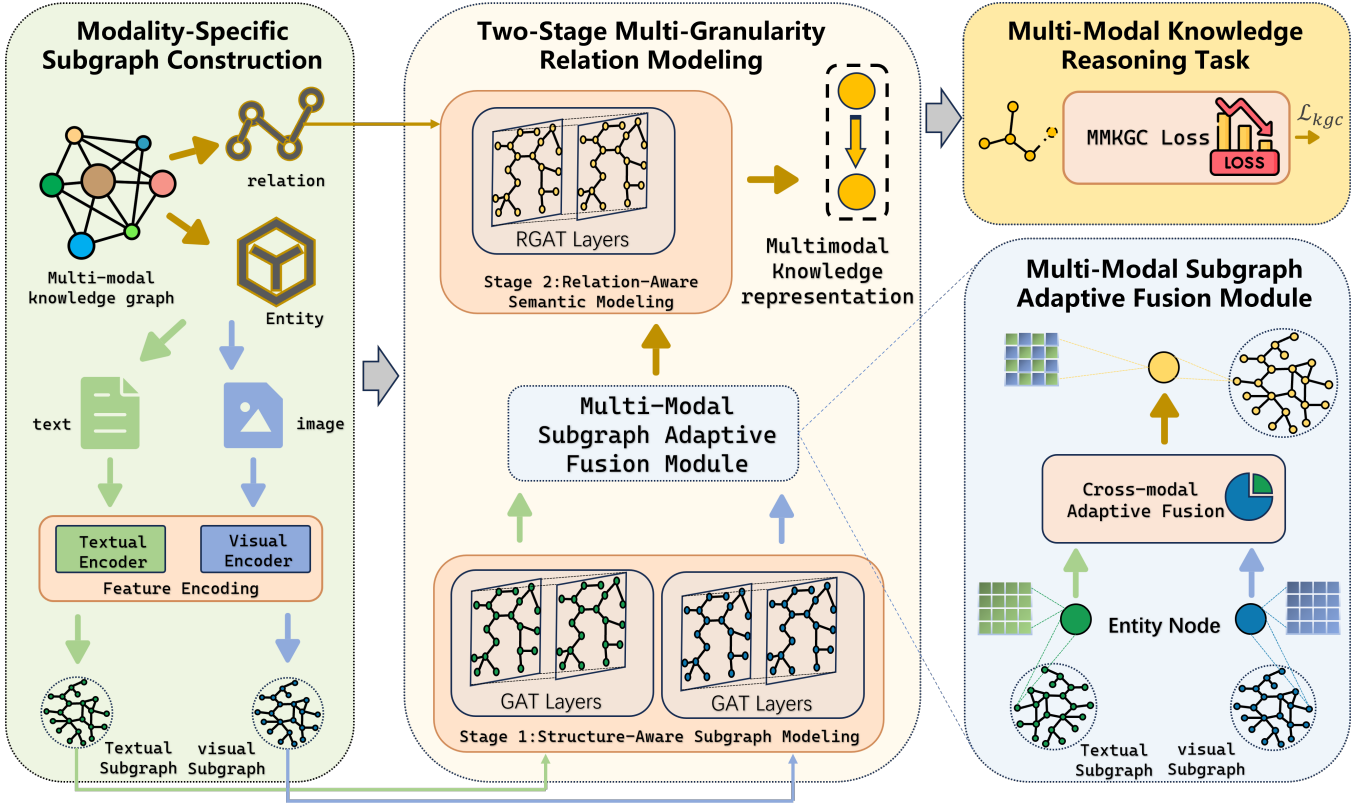


Figure 1: Overview of the SAFER framework. Our method addresses three key limitations in existing MMKG models by introducing (1) modality-specific subgraphs to preserve intra-modal structure, (2) multi-modal adaptive fusion to avoid early fusion conflicts, and (3) a two-stage multi-granularity relation modeling scheme that separates structural and semantic learning.

More recent advances focus on fine-grained multimodal alignment and structural integration. MyGO [35] discretizes image and text features into token sequences and performs hierarchical triple modeling. AdaMF-MAT [37] dynamically adjusts modality importance via adaptive fusion and employs modality-adversarial training to handle missing information. SGMPT [12] introduces structure-guided fusion into a multimodal transformer architecture, and HRGAT [13] builds hyper-node graphs to jointly reason over multimodal representations. Other methods such as MKGFormer [6], DRAGON [32], and NativE [36] further enhance multimodal representation learning with pretraining, topology-aware fusion, and robust fusion strategies.

Negative Sampling Approaches. Another direction enhances representation learning through adversarial or hierarchical negative sampling. KBGAN [3] introduces adversarial sampling to select informative negative triples, improving training efficiency. MM-RNS [30] applies multimodal hard negatives to strengthen learning signals. DHNS [19] proposes a diffusion-based hierarchical sampling strategy that exploits graph structure and modality-aware noise control to better train discriminative embeddings.

In contrast to existing work, our model SAFER constructs modality-specific subgraphs and performs adaptive fusion at the representation level, preserving graph topology while enabling

fine-grained semantic integration. We further incorporate relation-aware message passing in a multi-stage fashion to model hierarchical relational dependencies.

2.2 Multi-Modal Fusion Strategy

The goal of modality fusion strategies is to integrate information from various modalities (e.g., text, images, and audio) to produce more comprehensive and accurate entity and relationship representations. Based on the stage at which fusion occurs, these strategies are typically categorized into three types: early fusion, late fusion, and hybrid fusion[14].

Early fusion combines features from different modalities immediately after feature extraction, making it suitable for small datasets while enabling quick output generation and reducing the risk of overfitting. However, using fully connected layers for early fusion has certain limitations, such as requiring fixed input sizes and insufficient capability to capture complex inter-modal relationships. The MKGAT[21] model implements early fusion by employing a simple fully connected layer to learn features from multiple modalities, merging them into a dense vector to produce more suitable entity embeddings.

Late fusion, a commonly adopted method, processes features from each modality independently before combining them for

deeper analysis. This approach excels at capturing intricate relationships between modalities and handling missing data. In multi-modal knowledge graphs, optimizing energy functions is a core technique for late fusion. For example, IKRL[29] uses multiple TransE-based[2] scoring functions for modality interaction; RSME[26] employs gating mechanisms to selectively integrate modality-specific information; and OTKGE[4] leverages optimal transport methods for fusion. Additionally, models like EVA implement attention-weighted concatenation for late fusion.

Hybrid fusion combines the strengths of both early and late fusion, ensuring data consistency while leveraging complementary information across modalities. These strategies are widely used in reasoning tasks for multi-modal knowledge graphs, improving model performance and generalization capabilities. A representative hybrid fusion method is DFMKE[41], which uses concatenation for early fusion to ensure data consistency across modalities, while introducing a low-rank multi-modal fusion (LMF) module[16] for late fusion to effectively utilize complementary modality information. GraphAdapter [10] further enhances hybrid fusion by constructing dual knowledge graphs for text and image modalities, which are integrated via adapter modules in a vision-language model, preserving structure while enabling cross-modal alignment.

3 Methodology

This section introduces the overall architecture of our proposed model, **SAFER**, which learns expressive entity embeddings by integrating multi-modal semantics and structural relations in knowledge graphs. SAFER first constructs modality-specific subgraphs encoded via Graph Attention Networks (GAT), then dynamically aggregates these embeddings using an adaptive multi-modal fusion mechanism, and finally refines entity representations through a two-stage multi-granularity relation modeling approach. The overall framework is illustrated in Figure 1, with details presented in the following subsections.

3.1 Problem Definition

We define a multi-modal knowledge graph (MMKG) as a tuple $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T}, \mathcal{M})$, where $\mathcal{E}$ is the set of entities, $\mathcal{R}$ is the set of relations, and $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ is the set of relational triples. Each entity $e \in \mathcal{E}$ is associated with multiple modalities (e.g., textual descriptions, visual content), represented as a set of modality-specific features $\{\mathbf{h}_e^m \mid m \in \mathcal{M}\}$, where $\mathcal{M}$ denotes the modality set.

The goal of multi-modal knowledge representation learning is to learn a unified and expressive embedding $\mathbf{h}_e \in \mathbb{R}^d$ for each entity e , which captures both structural information from the knowledge graph and semantic content from multiple modalities. These embeddings are used to perform knowledge reasoning tasks such as link prediction, entity alignment, and knowledge completion.

3.2 Modality-Specific Subgraph Construction

We encode textual and visual features of entities using pretrained BERT and VGG16 models, respectively, and project these embeddings into a unified latent space:

$$\mathbf{h}_e^m = \mathbf{W}^m \cdot \text{Enc}^m(e), \quad m \in \{\text{text}, \text{image}\}. \quad (1)$$

Here, $\text{Enc}^m(e)$ denotes the raw feature vector of entity e in modality m , and $\mathbf{W}^m \in \mathbb{R}^{d \times d_m}$ is a trainable projection matrix that maps modality-specific features to a shared d -dimensional space.

To capture intra-modal structural dependencies, we construct modality-specific subgraphs using a shared adjacency matrix derived from the connectivity of original KG triples, irrespective of relation types:

$$\mathcal{A}_{ij} = \begin{cases} 1, & (e_i, r, e_j) \in \mathcal{T} \text{ or } (e_j, r, e_i) \in \mathcal{T} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Here, $\mathcal{A} \in \{0, 1\}^{|\mathcal{E}| \times |\mathcal{E}|}$ encodes the undirected connectivity among entities based on existing triples, ignoring relation types to focus on structural proximity.

Graph Attention Networks (GAT) independently encode these subgraphs, performing modality-sensitive aggregation of neighboring information:

$$\mathbf{h}_i^{m, \text{out}} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^m \mathbf{W} \mathbf{h}_j^m \right). \quad (3)$$

Here, $\mathbf{h}_i^{m, \text{out}}$ denotes the updated embedding of entity e_i in modality m , $\mathbf{W}$ is a shared transformation matrix, α_{ij}^m is the attention coefficient between entities e_i and e_j , and $\mathcal{N}(i)$ represents the neighbors of e_i in the shared adjacency graph.

This approach effectively preserves modality-specific local structures, facilitating subsequent adaptive multi-modal fusion and relational reasoning.

3.3 Multi-Modal Subgraph Adaptive Fusion

After obtaining modality-specific structural embeddings, we aim to integrate these heterogeneous features into unified entity representations. Previous methods typically fuse modalities by merging subgraphs directly at the topology level, often causing structural conflicts and semantic misalignment. To address these issues, we propose a multi-modal adaptive fusion strategy at the semantic level, which aggregates modality-specific embeddings into a unified semantic space. This approach preserves structural consistency, mitigates interference, and captures complementary semantic signals effectively, resulting in stable and expressive representations for downstream knowledge reasoning.

Multi-Modal Fusion Objective. Let $\mathbf{h}_e^{\text{text}}$ and $\mathbf{h}_e^{\text{image}}$ denote the representations of entity e obtained from the textual and visual subgraphs, respectively. Our goal is to generate a unified embedding $\mathbf{h}_e \in \mathbb{R}^d$ that effectively captures complementary modality-specific information in a shared semantic space.

Adaptive Multi-Modal Fusion. To enable flexible and entity-specific modality integration, we propose a multi-modal adaptive fusion mechanism that dynamically assigns attention weights across modalities. For each modality $m \in \{\text{text}, \text{image}\}$, we compute:

$$\alpha_e^m = \frac{\exp(\mathbf{q}^\top \cdot \mathbf{h}_e^m)}{\sum_{m' \in \mathcal{M}} \exp(\mathbf{q}^\top \cdot \mathbf{h}_e^{m'})} \quad (4)$$

where $\mathbf{q} \in \mathbb{R}^d$ is a learnable query vector shared across all entities. These attention weights α_e^m are normalized across modalities and vary by entity, enabling adaptive control over the contribution of

each modality. The final fused representation is then computed as:

$$\mathbf{h}_e = \sum_{m \in \mathcal{M}} \alpha_e^m \cdot \mathbf{h}_e^m \quad (5)$$

Unified Multi-Modal Graph Representation. The fused representations $\{\mathbf{h}_e\}_{e \in \mathcal{E}}$ are used as node features in a unified graph. The topology of this graph is inherited from the original entity connectivity used in the modality-specific subgraphs, ensuring structural consistency. This unified representation not only integrates multi-modal signals at the semantic level, but also preserves the relational structure necessary for the subsequent relation-aware semantic modeling.

3.4 Two-Stage Multi-Granularity Relation Modeling

After obtaining fused entity representations from the multi-modal subgraph fusion stage, we further enhance relational reasoning by explicitly modeling semantic interactions through relation-aware message passing. To achieve this, we design a two-stage strategy that decouples structural and semantic modeling, enabling the capture of relational dependencies at different levels of granularity.

Stage 1: Structure-Aware Representation Learning. In the first stage, we construct modality-specific subgraphs based on the connectivity of the original knowledge graph, where relation types are temporarily ignored. Each subgraph is encoded using Graph Attention Networks (GAT), producing structure-aware embeddings that preserve intra-modal topological features. These modality-specific embeddings are then fused via an attention-based mechanism to generate a unified entity representation $\mathbf{h}_e \in \mathbb{R}^d$. While this fused embedding incorporates multi-modal and structural information, it lacks explicit relational semantics because all edges are treated as untyped during this stage.

Stage 2: Relation-Aware Semantic Modeling. To inject relational semantics, we take the fused entity embeddings $\mathbf{h}_e$ as input node features and reintroduce the original triple structure (h, r, t) in the second stage. We adopt a Relational Graph Attention Network (RGAT) to conduct relation-aware message passing on the fused entity graph. Each relation type $r \in \mathcal{R}$ is associated with a learnable embedding $\mathbf{r}_r \in \mathbb{R}^d$. For each triple, the attention score from head entity h to tail entity t under relation r is computed as:

$$\alpha_{ht}^{(r)} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_h \parallel \mathbf{W}\mathbf{h}_t \parallel \mathbf{W}_r \mathbf{r}_r]))}{\sum_{t' \in \mathcal{N}_r(h)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_h \parallel \mathbf{W}\mathbf{h}_{t'} \parallel \mathbf{W}_r \mathbf{r}_r]))} \quad (6)$$

where $\mathcal{N}_r(h)$ denotes the set of tail entities connected to h via relation r , $\mathbf{W}$, $\mathbf{W}_r$ are projection matrices, and $\mathbf{a}$ is a shared attention vector.

The final entity representation is updated by aggregating relational messages:

$$\tilde{\mathbf{h}}_h = \sum_{r \in \mathcal{R}} \sum_{t \in \mathcal{N}_r(h)} \alpha_{ht}^{(r)} \cdot \mathbf{W}\mathbf{h}_t \quad (7)$$

We apply multi-head attention and stack multiple RGAT layers to capture diverse semantic dependencies. The resulting embeddings $\tilde{\mathbf{h}}_e$ integrate topological structure, modality-aware features, and fine-grained relational semantics, and are used in downstream knowledge reasoning tasks such as link prediction and triple scoring.

Algorithm 1: Training Procedure of SAFER

Input: Positive triples $\mathcal{T}^+$,
Modality features $\mathbf{X}^{\text{text}}, \mathbf{X}^{\text{image}}$,
Learning rate η , batch size B
Output: Trained SAFER model $\mathcal{M}_\theta$
while *not converged* **do**
 Sample mini-batch $\mathcal{B} \subset \mathcal{T}^+$;
 Generate negative triples $\mathcal{T}^-$ by corrupting h or t ;
 // Stage 1: Structure-aware representation learning
 Encode modality-specific subgraphs via GAT to obtain $\mathbf{h}_e^m$;
 Compute fused embedding $\mathbf{h}_e = \sum_m \alpha_e^m \cdot \mathbf{h}_e^m$;
 // Stage 2: Relation-aware semantic modeling
 Refine embeddings via RGAT to obtain $\tilde{\mathbf{h}}_e$;
 Compute triple scores with DistMult;
 Compute multi-modal knowledge graph completion loss based on softplus $\mathcal{L}_{\text{kgc}}$;
 Update parameters θ via Adam optimizer;
return Trained model $\mathcal{M}_\theta$

3.5 Multi-Modal Knowledge Reasoning and Training

We perform multi-modal knowledge reasoning for multi-modal knowledge graph completion (MMKGC) and link prediction via a triple scoring module built upon the final entity and relation embeddings produced by our SAFER framework. Specifically, we adopt the DistMult [31] scoring function to assess the plausibility of a triple (h, r, t) . Given the entity embeddings $\tilde{\mathbf{h}}_h, \tilde{\mathbf{h}}_t \in \mathbb{R}^d$ and relation embedding $\mathbf{r}_r \in \mathbb{R}^d$, the plausibility is calculated as:

$$f(h, r, t) = \sum_{i=1}^d \tilde{\mathbf{h}}_h^{(i)} \cdot \mathbf{r}_r^{(i)} \cdot \tilde{\mathbf{h}}_t^{(i)}. \quad (8)$$

To support bi-directional knowledge reasoning, we compute scores for both head and tail prediction:

$$f_{\text{head}}(h, r, t) = f(t, r^{-1}, h), \quad f_{\text{tail}}(h, r, t) = f(h, r, t), \quad (9)$$

and obtain the final triple plausibility score by averaging:

$$\text{Score}(h, r, t) = \frac{f_{\text{head}}(h, r, t) + f_{\text{tail}}(h, r, t)}{2}. \quad (10)$$

We optimize a softplus-based loss to distinguish valid triples $\mathcal{T}^+$ from corrupted ones $\mathcal{T}^-$:

$$\begin{aligned} \mathcal{L}_{\text{kgc}} = & \sum_{(h, r, t) \in \mathcal{T}^+} \log(1 + \exp(-f(h, r, t))) \\ & + \sum_{(h', r, t') \in \mathcal{T}^-} \log(1 + \exp(f(h', r, t'))) \end{aligned} \quad (11)$$

The training pipeline consists of: (1) modality-specific subgraph structure encoding; (2) multi-modal adaptive fusion to generate unified entity representations; (3) relational semantics modeling; and (4) scoring and loss calculation for triple plausibility. All parameters are optimized jointly using the Adam optimizer. The full training process of SAFER is summarized in Algorithm 1.

Table 1: Model performance comparison on DB15K and MKG-W datasets

Model	DB15K				MKG-W			
	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
Uni-modal KGC Models								
TransE[2] (NeurIPS 2013)	24.86	12.78	31.48	47.07	29.19	21.06	33.20	44.23
DistMult[31] (ICLR 2015)	23.03	14.78	26.28	39.59	20.99	15.93	22.28	30.86
ComplEx[24] (ACL 2016)	27.48	18.37	31.57	45.37	24.93	19.09	26.69	36.73
RotatE[22] (ICLR 2019)	29.28	17.87	36.12	49.66	33.67	26.80	36.68	46.73
PairRE[5] (ACL 2021)	31.13	21.67	35.91	49.30	34.40	28.24	36.71	46.04
GC-OTE[23] (ACL 2020)	31.85	22.11	36.52	51.18	33.92	26.55	35.96	46.05
Tucker[1] (ICML 2019)	33.86	25.33	37.91	50.38	30.39	24.44	32.91	41.25
Multi-modal KGC Models								
IKRL[29] (IJCAI 2017)	26.82	14.09	34.93	49.09	32.36	26.11	34.75	44.07
TBKGC[18] (NAACL 2018)	28.40	15.61	37.03	49.86	31.48	25.31	33.98	43.24
TransAE[28] (IJCNN 2019)	28.09	21.25	31.17	41.17	30.00	21.23	34.91	44.72
MMKRL[17] (APIN 2025)	26.81	13.85	35.07	49.39	30.10	22.16	34.09	44.69
RSME[26] (MM 2021)	29.76	24.15	32.12	40.29	29.23	23.36	31.97	40.43
VBKGC[38] (KDD 2022)	30.61	19.75	37.18	49.44	30.61	24.91	33.01	40.88
OTKGE[4] (NeurIPS 2022)	23.86	18.45	25.89	34.23	34.36	28.85	36.25	44.88
IMF[11] (WWW 2023)	32.25	24.20	36.00	48.19	34.50	28.77	36.62	45.44
QEB[27] (MM 2023)	28.18	14.82	36.67	51.55	32.38	25.47	35.06	45.32
VISTA[8] (EMNLP 2023)	30.42	22.49	33.56	45.94	32.91	26.12	35.38	45.61
AdaMF[37] (COLING 2024)	32.51	21.31	39.67	51.68	34.27	27.21	37.86	47.21
MyGO[35] (AAAI 2025)	<u>37.72</u>	<u>30.08</u>	41.26	52.21	<u>36.10</u>	<u>29.78</u>	38.54	47.75
Negative Sampling Models								
KBGAN[3] (NAACL 2018)	25.73	9.91	36.95	51.93	29.47	22.21	34.87	40.64
MANS[34] (IJCNN 2023)	28.82	16.87	36.58	49.26	30.88	24.89	33.63	41.78
MMRNS[30] (MM 2022)	32.68	23.01	37.86	51.01	35.03	28.59	37.49	47.47
DHNS[19] (DASFAA 2025)	34.36	23.34	<u>41.56</u>	<u>53.72</u>	35.68	28.98	<u>38.68</u>	<u>48.12</u>
SAFER-TransE	33.11	26.87	35.95	46.10	35.24	28.81	37.83	46.32
SAFER-RotatE	34.45	27.14	37.27	47.85	36.03	29.33	38.70	47.93
SAFER-ComplEx	34.88	24.76	37.90	48.10	36.24	29.76	38.95	48.20
SAFER-DistMult(ours)	36.35	27.33	41.30	53.00	36.77	29.99	39.72	48.84

Table 2: Statistical information of the datasets.

Dataset	Entity	Relation	#Train	#Valid	#Test	Image	Text
DB15K[15]	12,842	279	79,222	9,902	9,904	12818	9078
MKG-W[30]	15,000	169	34,196	4,276	4,274	14463	14123

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate SAFER on two real-world multi-modal knowledge graph datasets: **DB15K**[15] and **MKG-W**[30]. Both datasets provide structural triples along with entity-level textual and visual features. DB15K is a subset of DBpedia[9] annotated with multimodal content, while MKG-W is constructed from Wikidata[25] and aligned with Open Images and Wikipedia descriptions. Table 2 summarizes the statistics of the two datasets.

Evaluation Metrics. To comprehensively assess the performance of our model on the link prediction task, we adopt a set of standard and widely recognized evaluation metrics. This task is treated as a ranking problem: for a given test triple (h, r, t) , the model is asked to rank the correct tail entity t against a set of candidate entities (or similarly for the head entity h). Our evaluation is based on the rank of the correct entity. The primary metrics used are Mean Reciprocal Rank (MRR) and Hits@K for $K \in \{1, 3, 10\}$.

Baselines. We compare SAFER against a wide range of baseline models, categorized into three groups:

(1) **Uni-modal KGC Models:** TransE [2], DistMult [31], ComplEx [24], RotatE [22], PairRE [5], GC-OTE [23], and Tucker [1]. These models rely purely on structural triples and do not incorporate external modalities.

(2) **Multi-modal KGC Models:** IKRL [29], TBKGC [18], TransAE [28], MMKRL [17], RSME [26], VBKGC [38], OTKGE [4], IMF [11], QEB [27], VISTA [8], AdaMF [37], and MyGO [35]. These

Table 3: Ablation study across fusion strategies, modality contribution, and relation modeling methods.

Variant	Setting	DB15K				MKG-W			
		MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
Modality Fusion Strategy									
SAFER (Full)	Adaptive Fusion	36.35	27.33	41.30	53.00	36.77	29.99	39.72	48.84
Average Fusion	Average	34.51	25.21	39.05	51.08	35.12	28.02	37.21	46.93
Concat Fusion	Concatenation	34.07	24.93	38.41	50.62	34.86	27.85	36.89	46.51
DotProduct Fusion	Dot Product	34.82	25.71	39.43	51.32	35.43	28.41	37.83	47.10
Modality Contribution									
SAFER (Full)	Multimodal	36.35	27.33	41.30	53.00	36.77	29.99	39.72	48.84
Text-only	Text Modality Only	32.98	23.51	37.15	48.74	33.77	26.10	35.91	44.80
Image-only	Image Modality Only	31.76	22.48	35.37	47.21	32.66	25.45	34.21	43.72
Structure and Semantic Modeling									
Structure+Semantics (Full)	Two-Stage	36.35	27.33	41.30	53.00	36.77	29.99	39.72	48.84
Structure Only	Single-Stage	33.99	24.19	38.75	50.85	34.62	27.50	37.29	46.81
Semantics Only	Single-Stage	32.45	22.77	36.88	48.03	33.45	25.77	35.68	44.21
Structure+Structure	Two-Stage	34.01	24.62	38.32	50.19	34.80	27.61	36.80	46.30
Semantics+Semantics	Two-Stage	33.32	23.90	37.54	49.36	34.03	26.83	36.05	45.23

methods enhance structural knowledge by incorporating image and text features associated with entities.

(3) Negative Sampling Models: KBGAN [3], MANS [34], MM-RNS [30], and DHNS [19], which improve model learning through advanced sampling techniques or adversarial training strategies.

Implementation Details. Our SAFER model is implemented using PyTorch and the DGL framework. All experiments are conducted on a Linux server equipped with a single NVIDIA A100 GPU. We set the embedding dimension to 250 and use a batch size of 1024. The training is performed for 800 epochs using the Adam optimizer. The model architecture includes two GAT layers for structural encoding and two RGAT layers for semantic relation modeling. Baseline models are reproduced following their official implementations whenever possible, and some results are referred from MyGO [35].

4.2 Overall Performance Comparison

We evaluate SAFER against a diverse set of state-of-the-art baselines on the DB15K and MKG-W datasets. Table 1 presents the results under standard link prediction metrics: MRR, Hits@1, Hits@3, and Hits@10. On MKG-W, SAFER achieves the best overall performance with an MRR of 36.77%, Hits@1 of 29.99%, and Hits@10 of 48.84%, outperforming the strongest baseline MyGO [35]. Specifically, SAFER yields absolute improvements of +0.71% in Hits@1, +1.50% in Hits@10, and +2.69% in Hits@3. These gains demonstrate SAFER’s superior ability in precise entity ranking and overall link prediction quality. SAFER also shows strong results on DB15K, reaching an MRR of 36.35% and Hits@10 of 53.00%, which is competitive with the best-performing models under both the multi-modal and negative sampling categories, including MyGO and DHNS.

Compared to uni-modal KGE methods (e.g., TransE [2], DistMult [31], RotatE [22]), SAFER benefits significantly from the integration of textual and visual modalities, enabling richer semantic representations.

While existing multi-modal models like AdaMF [37] and MyGO [35] also incorporate auxiliary information, their fusion strategies are often static or early-stage. SAFER’s modality-specific

subgraph encoding and adaptive fusion mechanism enable entity-specific and dynamically weighted integration, effectively enhancing semantic alignment and reducing noise.

Furthermore, SAFER surpasses negative sampling-based models such as MMRNS [30] and DHNS [19], indicating that its improvements are achieved through better representation learning rather than external sampling heuristics.

In addition, the bottom section of Table 1 reports the performance of SAFER under different KGC scoring functions—TransE [2], DistMult [31], RotatE [22], and ComplEx [24]—on both datasets. These results show that SAFER provides consistent performance gains regardless of the decoder choice, with DistMult achieving the highest overall results. We therefore adopt DistMult as the default scoring function in all other experiments.

4.3 Ablation Study

To rigorously validate our design choices and dissect the contribution of each key component within the SAFER architecture, we conduct a comprehensive series of ablation studies on both datasets. This dual-dataset evaluation allows us to assess the robustness and generalizability of our design choices across knowledge graphs with different scales, structural properties, and modality characteristics. The results are shown in Table 3.

Modality Fusion Mechanism. We investigate the effectiveness of our proposed adaptive fusion strategy. We compare it against three widely-used alternatives: **average fusion**, **concatenation fusion**, and **dot-product-based fusion**. As evidenced by the results on both DB15K and MKG-W, replacing our adaptive fusion with any of these simpler strategies leads to a consistent and significant decline in performance across all metrics. Specifically, the static nature of average and concatenation fusion fails to account for entity-specific modality importance, a limitation observed on both datasets. While dot-product-based fusion offers a slight improvement, its performance remains inferior to our more expressive adaptive mechanism. This outcome, consistently observed across two distinct datasets, strongly verifies that a fine-grained, context-aware adaptive control is a generally effective strategy for multi-modal knowledge graph reasoning.

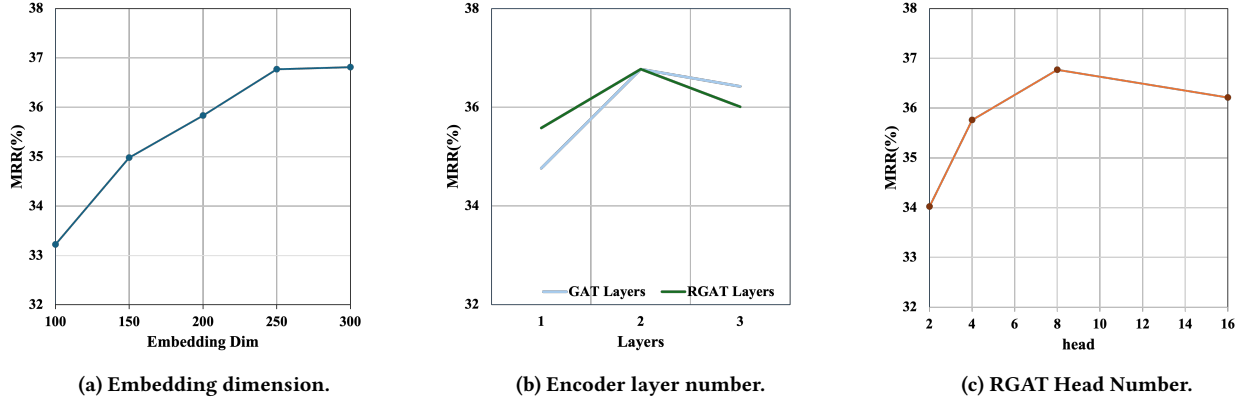


Figure 2: Parameter sensitivity analysis under different hyperparameter settings.

Modality Contribution.

To quantify the impact of each modality, we evaluate text-only and image-only variants of our model. Results show that the full SAFER model, leveraging synergistic modalities, achieves the best performance; removing either modality leads to a significant decline, confirming their complementarity. Furthermore, the text-only variant consistently outperforms the image-only one, indicating that text provides more foundational semantic signals. However, the significant gain from incorporating images highlights their critical role in enriching entity representations and providing complementary information.

Structure and Semantic Modeling. we evaluate the effectiveness of SAFER’s multi-granularity relation modeling, which decouples topological encoding from relational semantic reasoning. We test three variants: (1) a model with the **structural encoder disabled**, (2) a model with the **semantic encoder disabled**, and (3) a model using a **uniform relation modeling** scheme. The findings are highly consistent across both DB15K and MKG-W. Disabling either the structural or the semantic encoder leads to a substantial performance degradation, highlighting that both graph connectivity patterns and nuanced relational semantics are indispensable. These results, observed across datasets with different structural complexities, highlight the general necessity of our decoupled design to effectively alleviate the semantic under-specification problem in complex knowledge graphs.

4.4 Parameter Sensitivity

To rigorously assess the stability and generalization capability of SAFER, we conduct a comprehensive sensitivity analysis on its key hyperparameters. This analysis investigates how the model’s performance on the MKG-W dataset, measured by MRR, responds to variations in the Embedding Dimension, Encoder Depth, and Number of RGAT Layers Attention Heads. By systematically varying each of these parameters across a predefined range while holding others constant, we aim to demonstrate that SAFER’s strong performance is robust and not contingent on a narrow, finely-tuned configuration, thus confirming its practical applicability.

Embedding Dimension. We vary the embedding size $d \in \{100, 150, 200, 250, 300\}$ to investigate its influence on performance.

In Figure 2a, SAFER’s MRR steadily improves as the dimension increases, with a significant gain up to 250 dimensions. Beyond this point, the performance saturates. Hence, we adopt $d = 250$ as a balanced setting for efficiency and effectiveness.

Encoder Depth. We explore the effect of the number of layers in both GAT and RGAT modules by testing 1, 2, and 3-layer configurations. In Figure 2b, two layers achieve the best results. A single layer lacks sufficient capacity to model neighborhood information, while deeper networks (three layers) exhibit a slight performance drop due to potential over-smoothing or overfitting.

RGAT Head Number. We evaluate the impact of different numbers of attention heads in the RGAT encoder, selected from the set $\{2, 4, 8, 16\}$. As shown in Figure 2c, the model achieves the best performance with 8 attention heads. Using fewer heads (e.g., 2) limits the model’s capacity to capture diverse relational patterns, while using more heads (e.g., 16) introduces additional parameters and noise, which may slightly degrade generalization. Therefore, we set the number of attention heads to 8 as the default configuration.

5 Conclusion

In this paper, we propose a Multi-Granularity Multi-Modal Knowledge Graph Representation Learning Method via **Subgraph-aware Adaptive Fusion and Hierarchical Relation Modeling (SAFER)**, a novel framework for multi-modal knowledge graph representation learning. SAFER combines modality-specific subgraph construction, adaptive multi-modal fusion, and multi-granularity relation modeling to integrate heterogeneous features while preserving relational semantics. Experiments on DB15K and MKG-W show that SAFER consistently outperforms strong baselines across multiple metrics. Additional ablation and sensitivity studies validate its robustness and generalization. Future work includes extending SAFER to more modalities (e.g., audio, video) and applying it to complex tasks like multi-hop reasoning and zero-shot entity prediction.

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